

Thread - 2023-12-03

<https://x.com/RiikonenMarko/status/1731386591975821425>

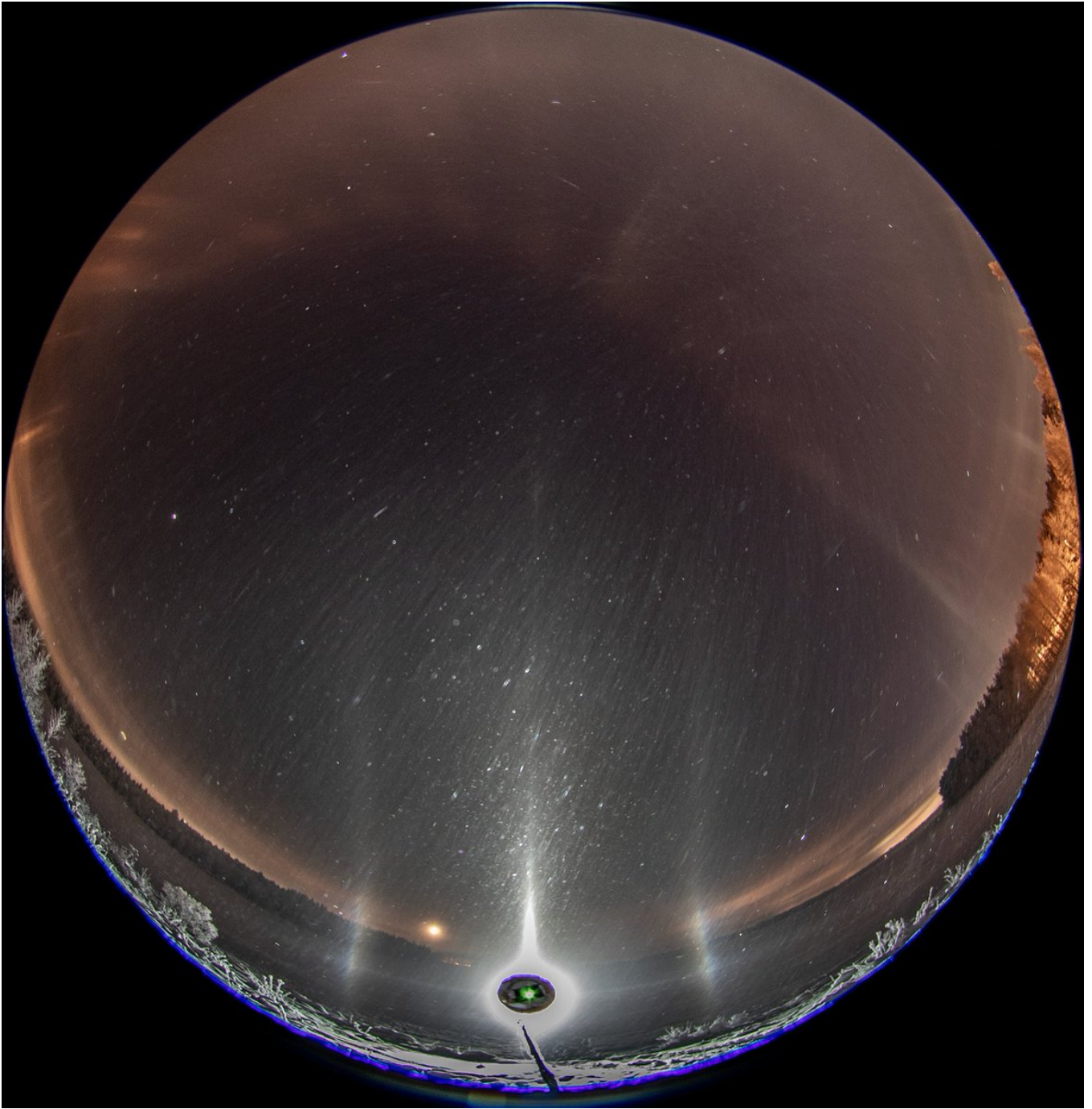
1/6 - 2023-12-03 18:55 UTC

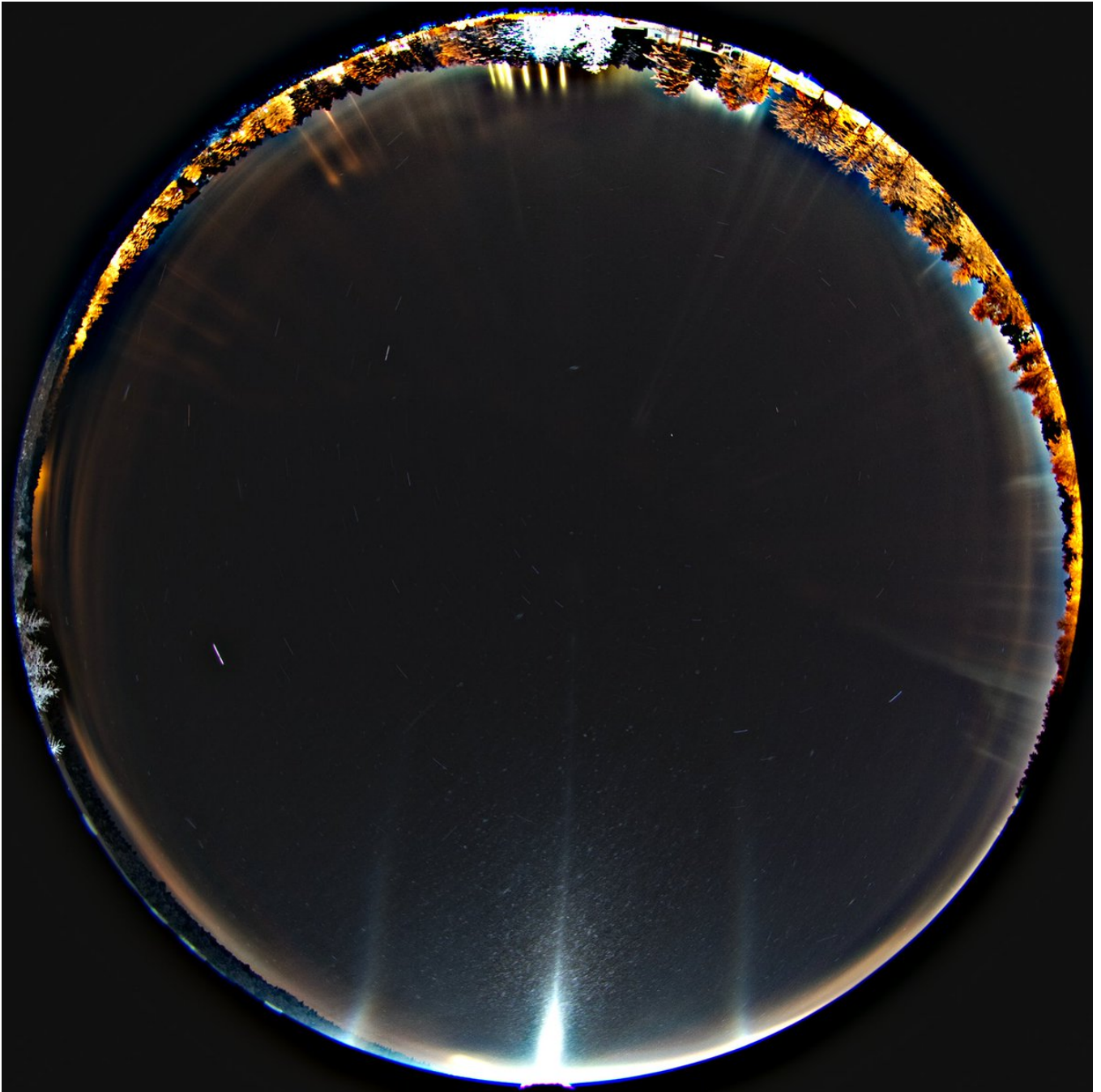
2023 Nov 20/21 - stage 3 - 1/6. After stage 1 and 2 (earlier posts) ice fog vanished from my spot at the east end of lake Salmijärvi. I was thinking of decamping but decided to wait. That was for the good: ice fog came back with these long subparhelic arcs and the display composition was stable, meaning crystal sample is representative. I had not tightened enough the tripod head which is why it was sliding. Temp -27 C in my own thermometer at about standing height.

[video: 1731386591975821425-GAcceeFWQAAlnvB.mp4]

2/6 - 2023-12-03 19:06 UTC

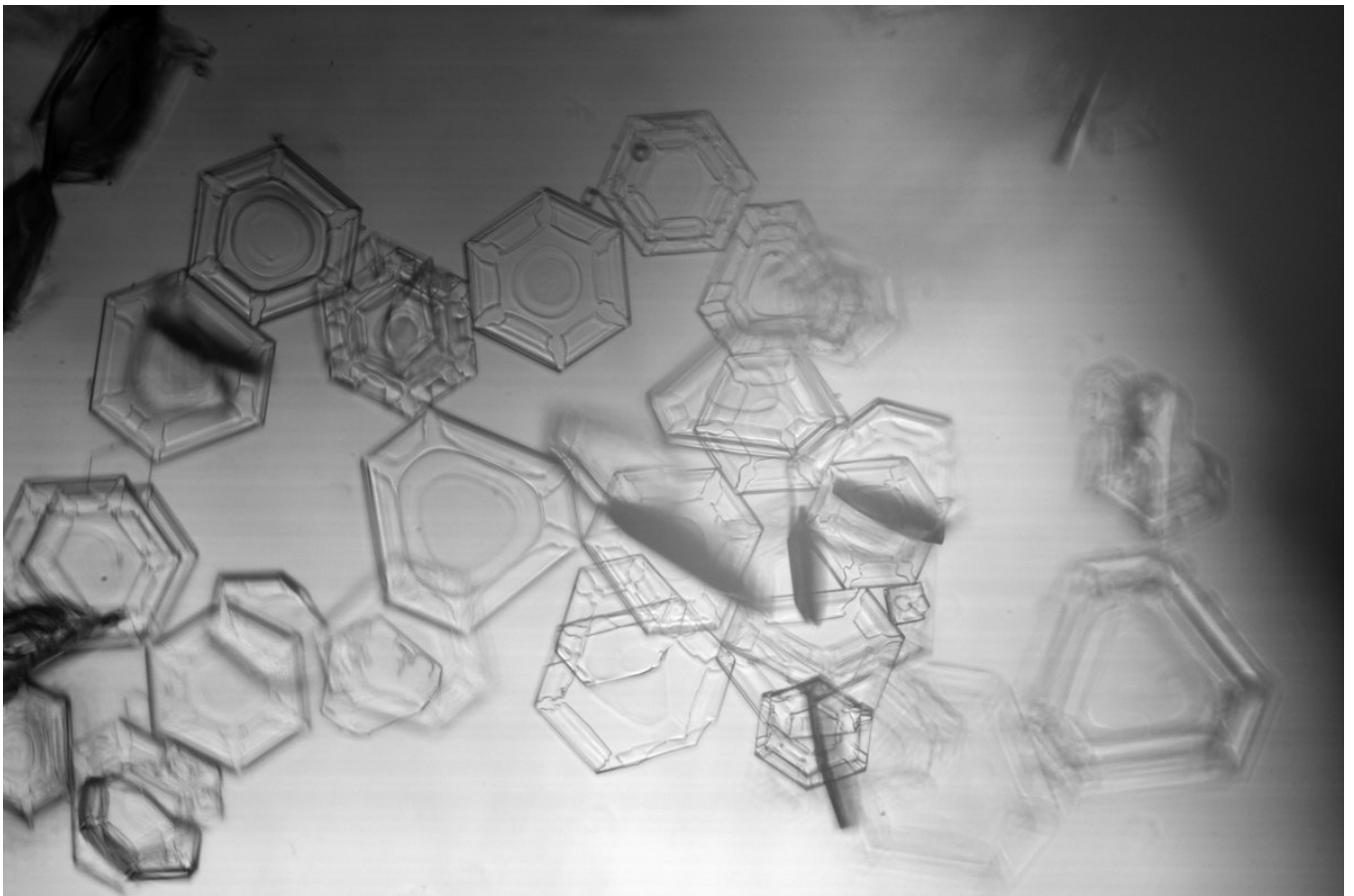
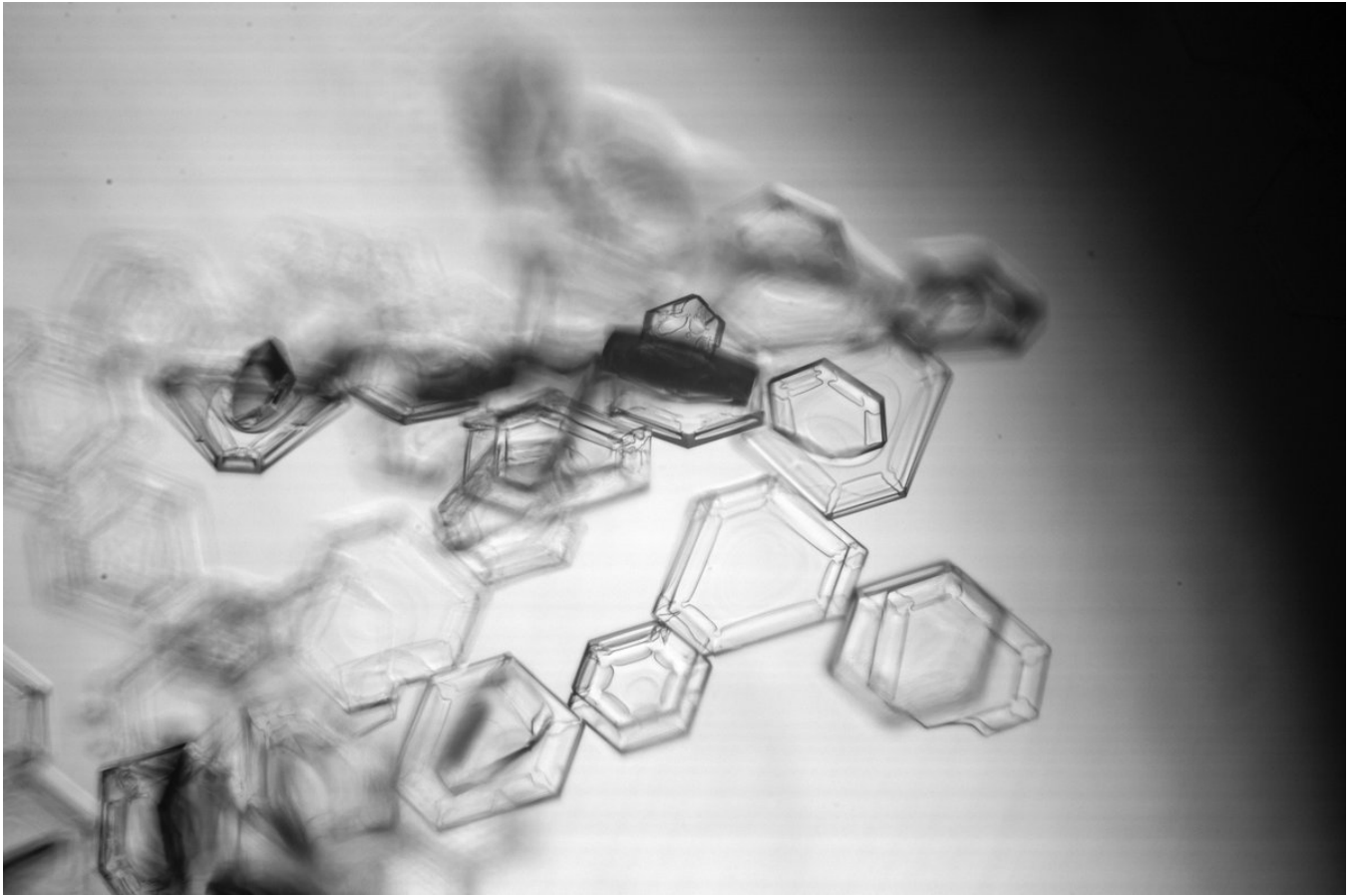
2023 Nov 20/21 - stage 3 - 2/6. The image on the left is the first frame I got, the other is a stack of 20 or so frames. Dish was put out right after the first frame. As the display vanished away without change, there was no pollution during crystal photography, which started 10 minutes later.

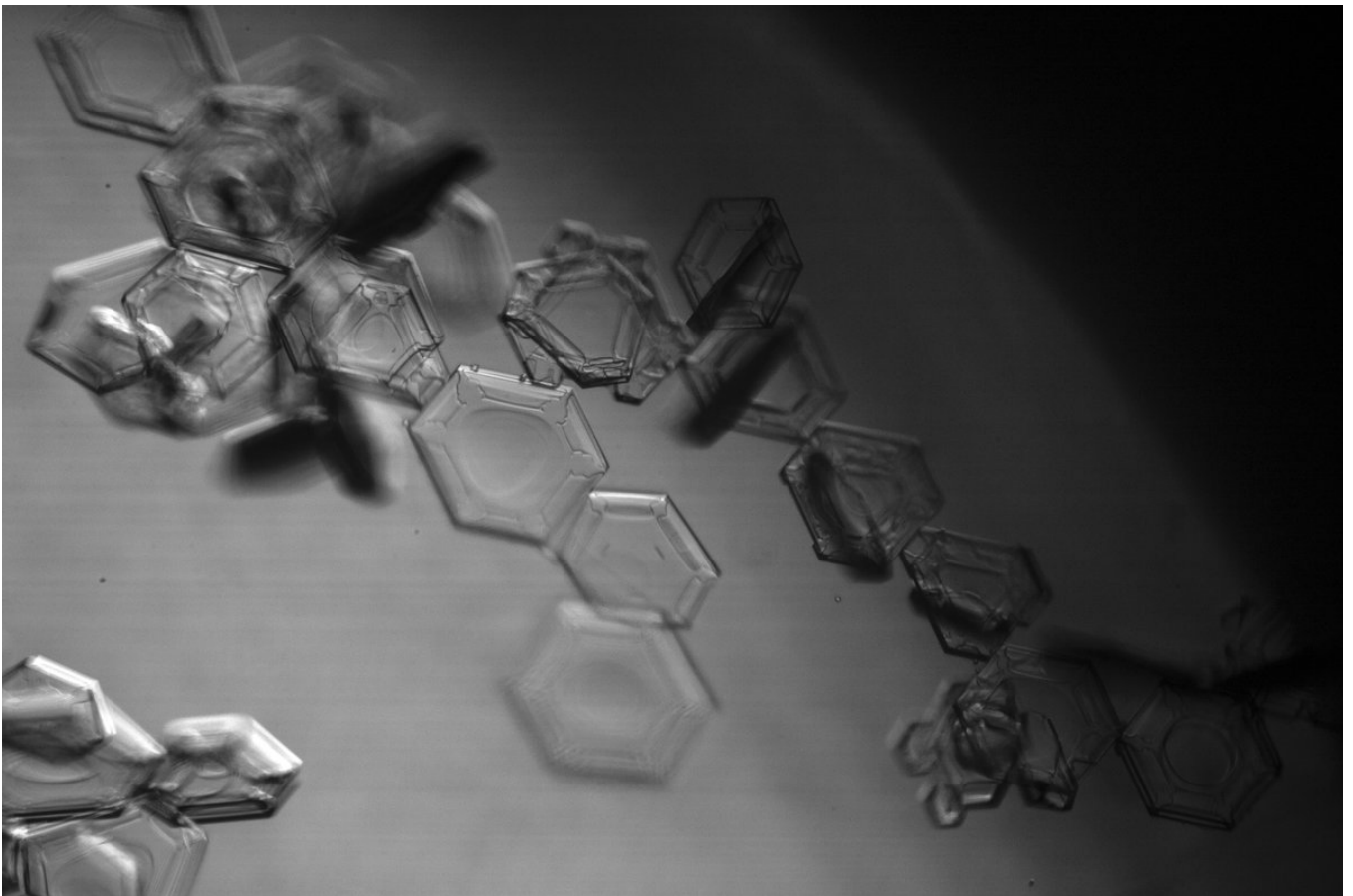
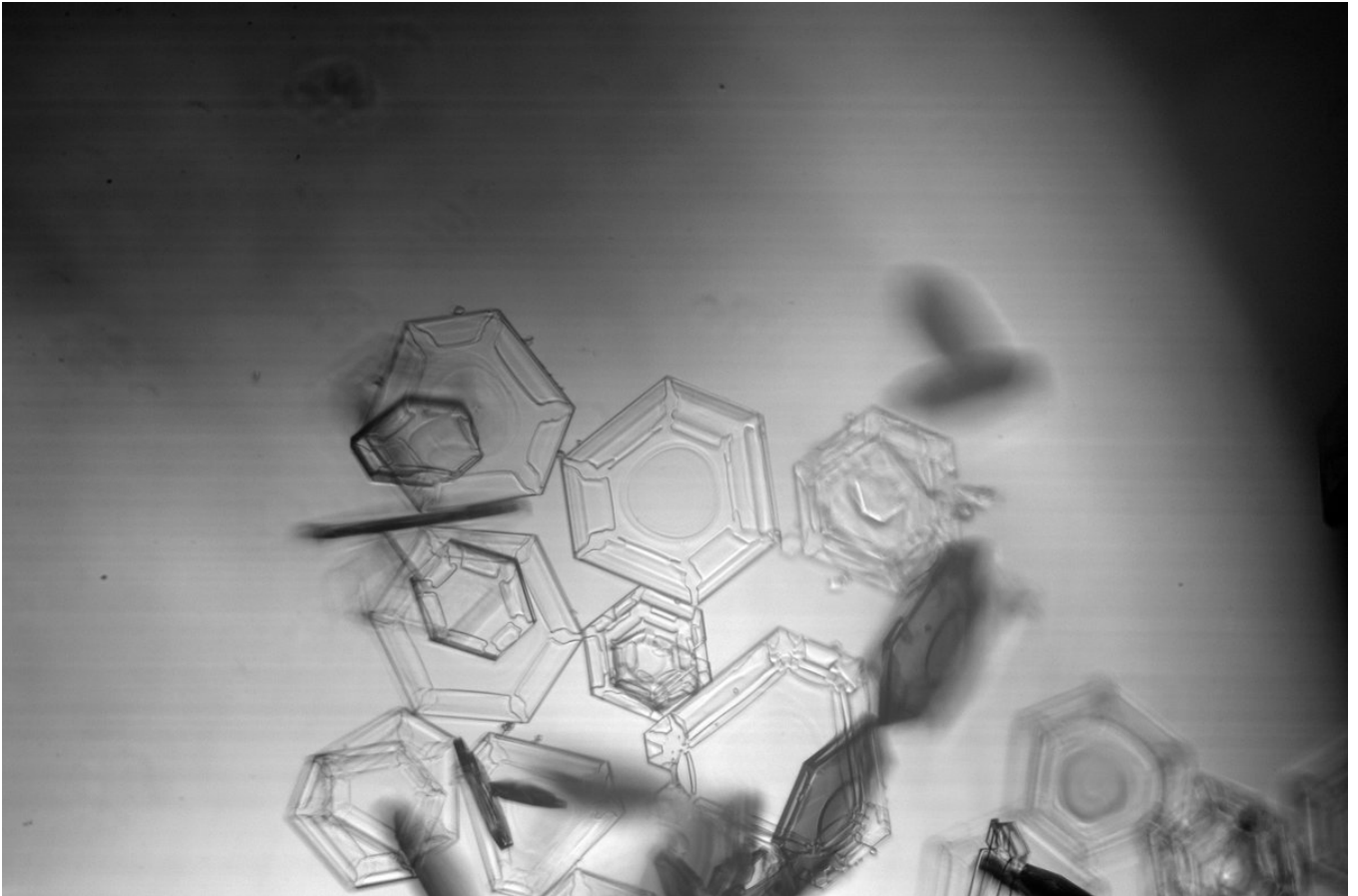




3/6 - 2023-12-03 19:14 UTC

2023 Nov 20/21 - stage 3 - 3/6. The sample is quite different from stages 1 and 2. It is distinctly uniform, filled with these big decorated plates. Interestingly, there is no halos in the display from light passing through 90 crystal angle.



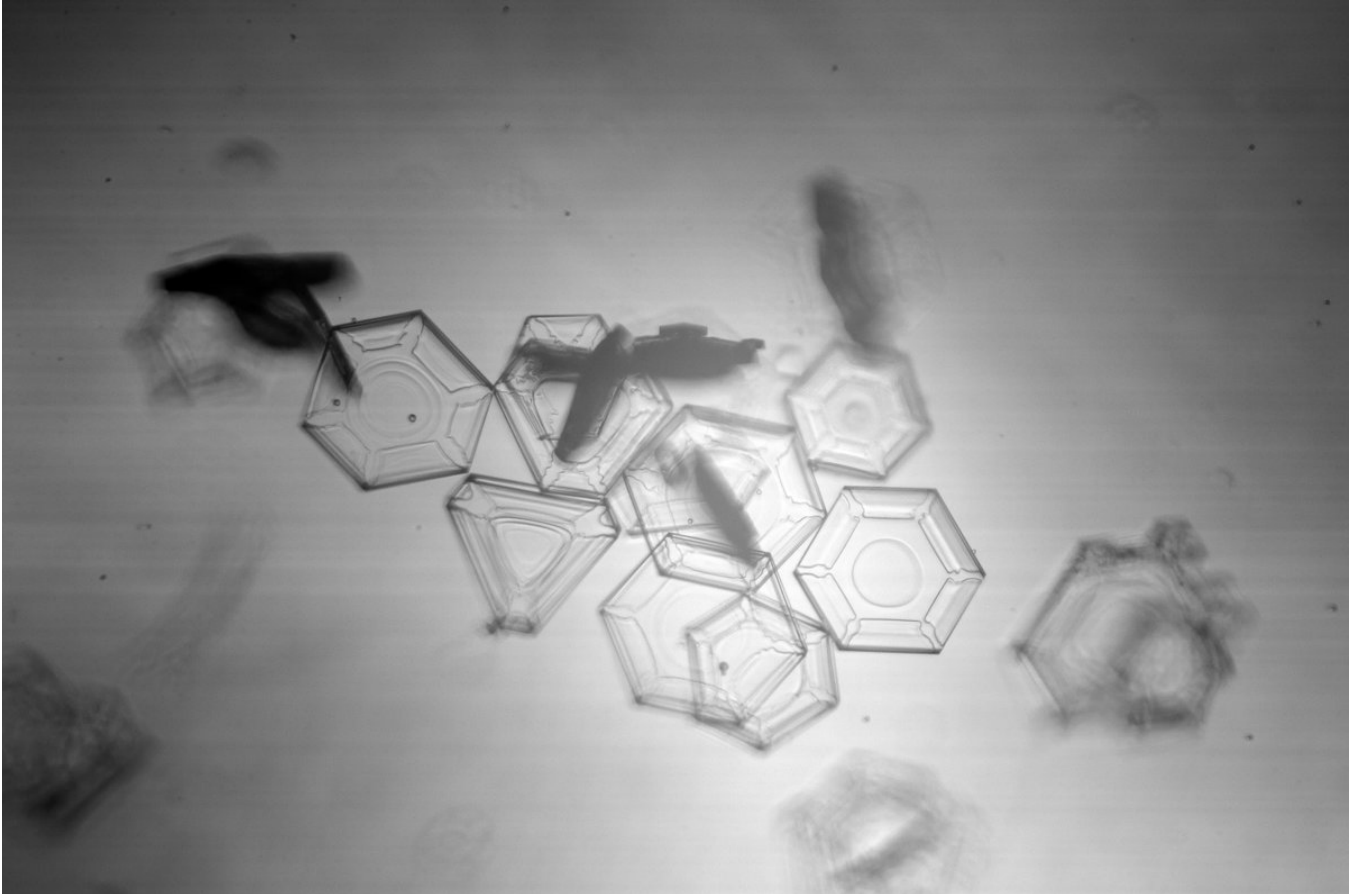


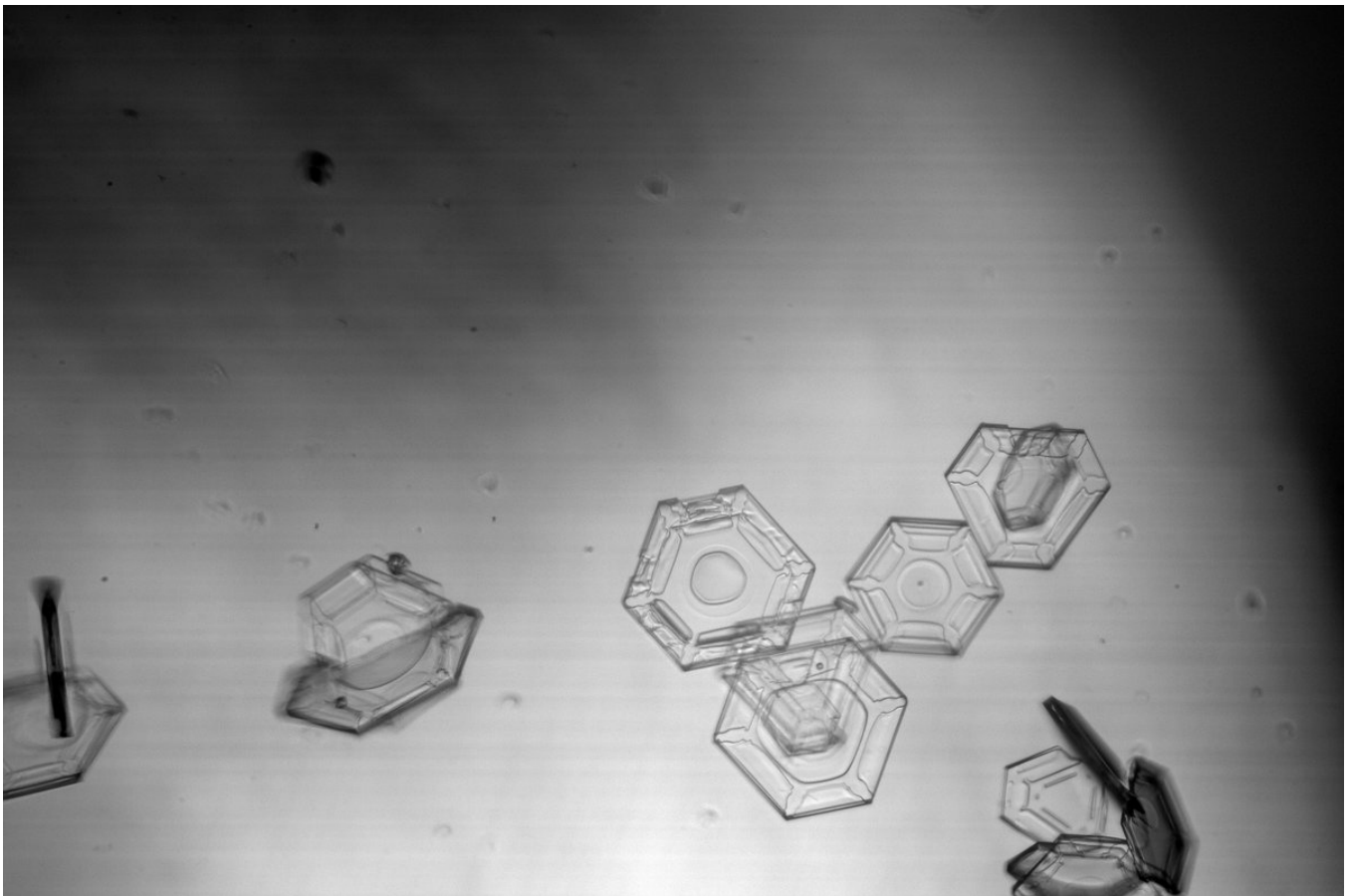
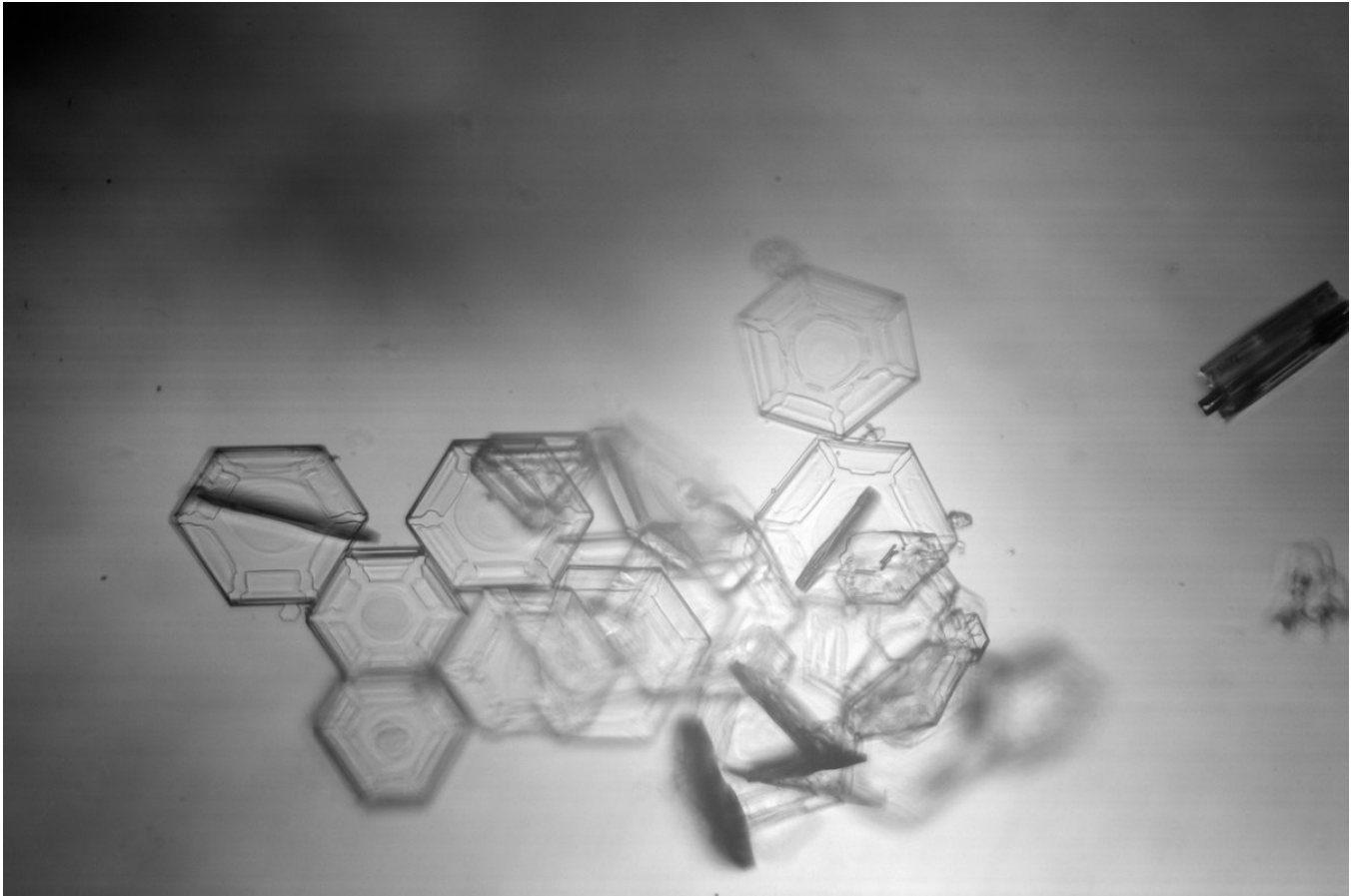
4/6 - 2023-12-03 19:25 UTC

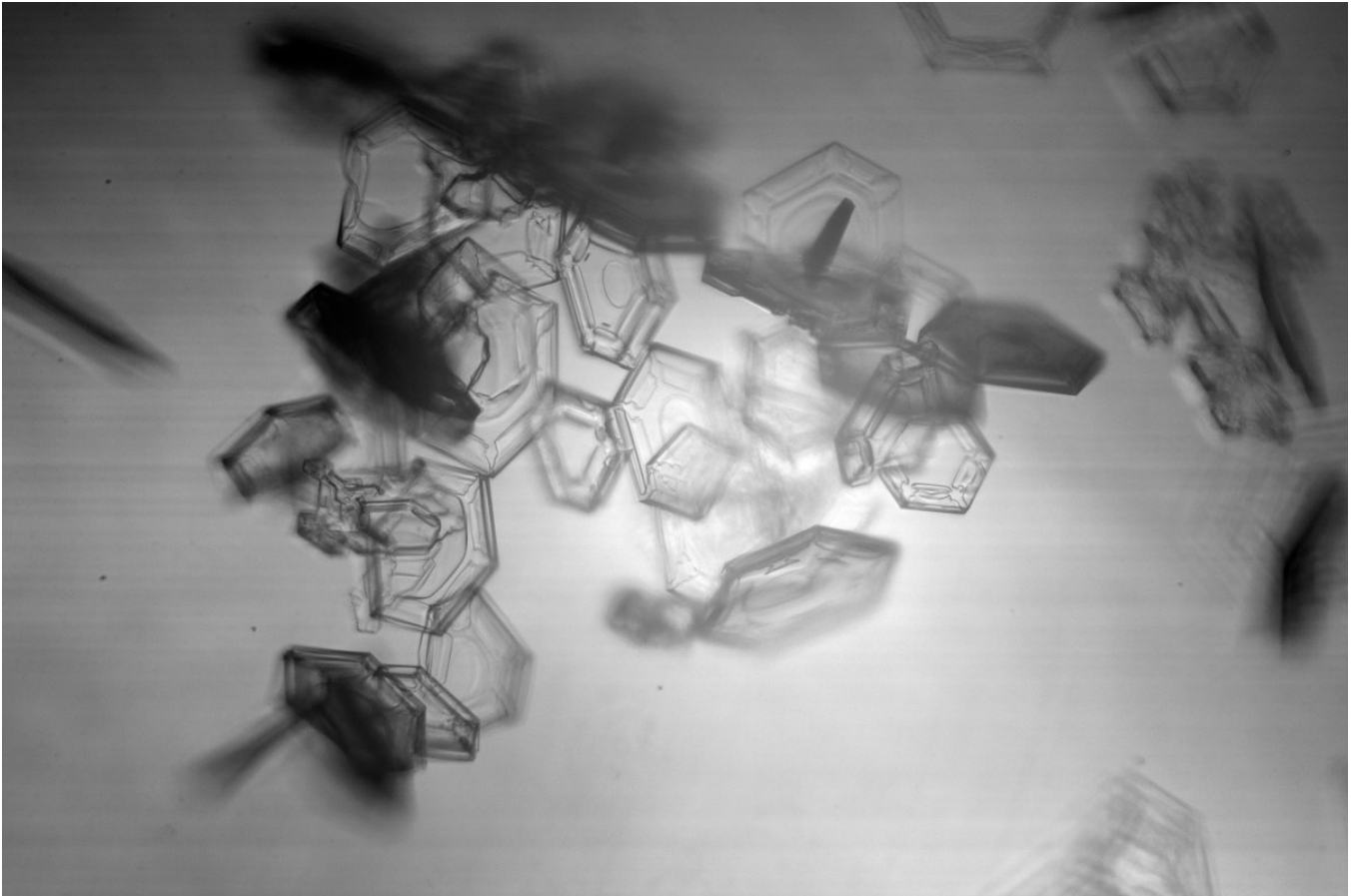
2023 Nov 20/21 - stage 3 - 4/6. More crystals. I found an interesting discussion on crystal sampling in

the comments of this blog post: <https://t.co/eXQBIAmxcF>

The answer is that, save for perhaps the most wildly changing displays, well organized one guy can do it all, photograph the halos and have crystal samples targeted to different stages of the display. I have also devised a simple protection to avoid sample pollution during photography. It is a cardboard plate with a hole for microscope objective. You can completely seal the dish this way. That's at least the theory, still waiting for the field trial.



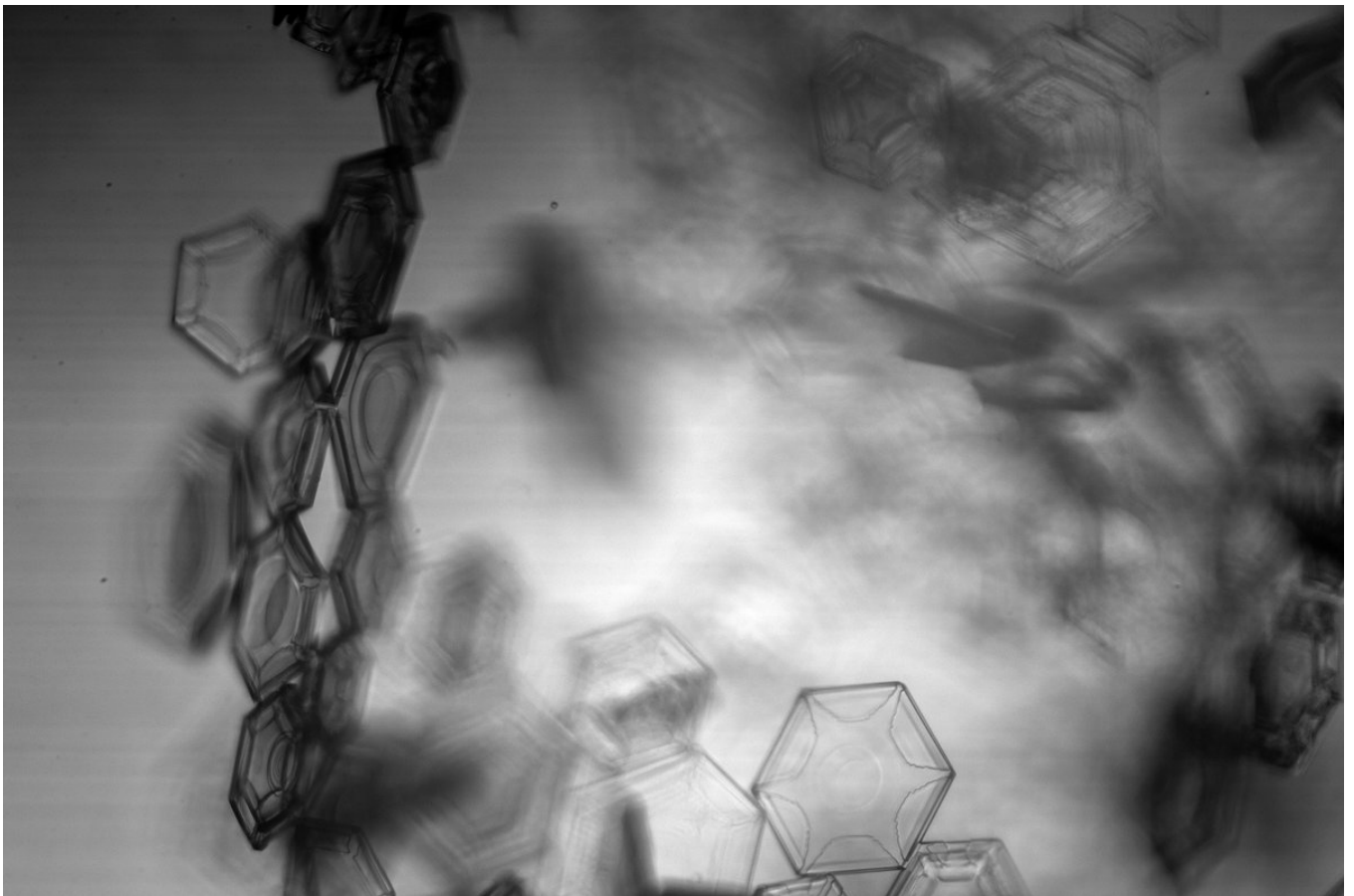
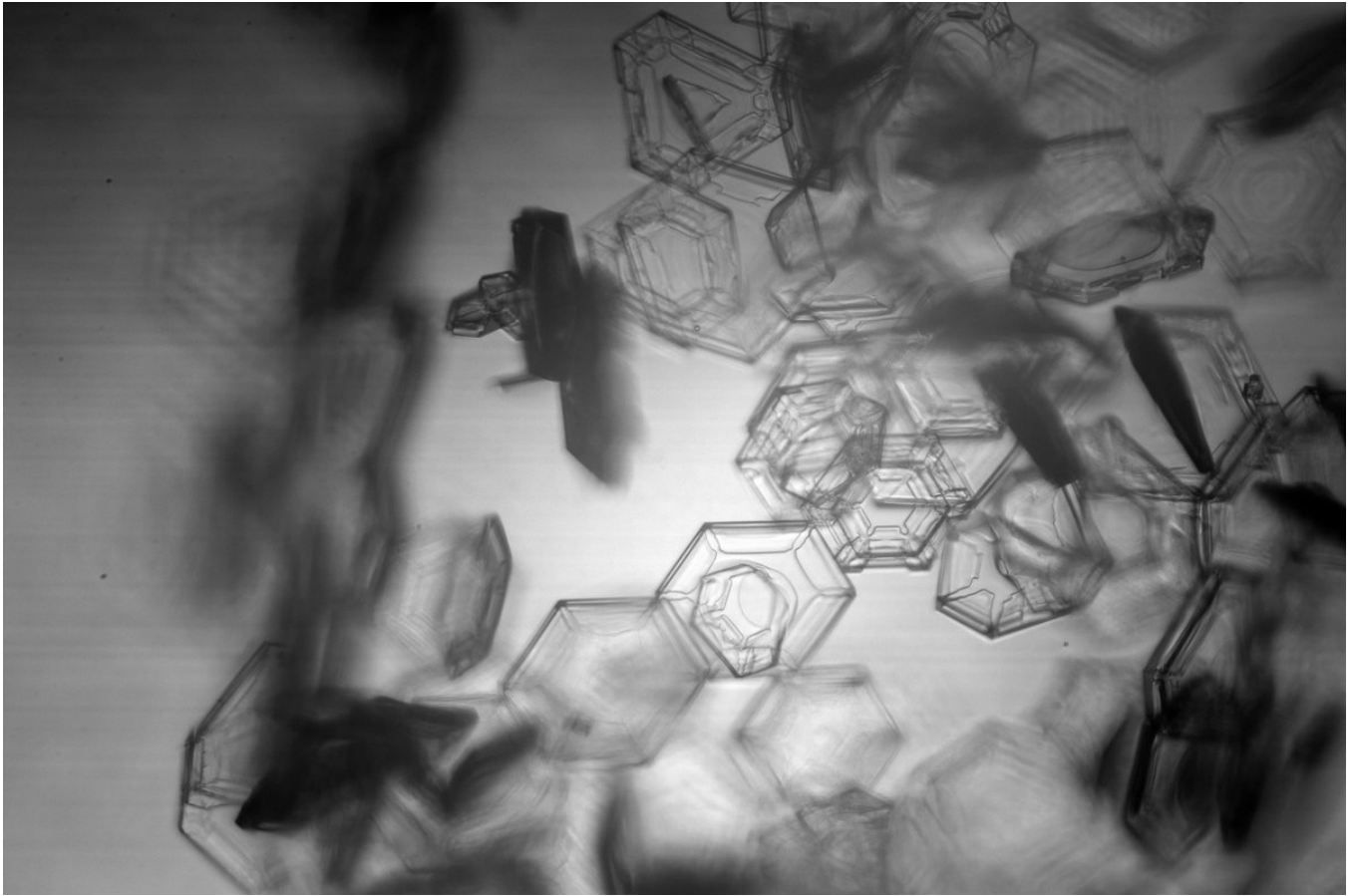


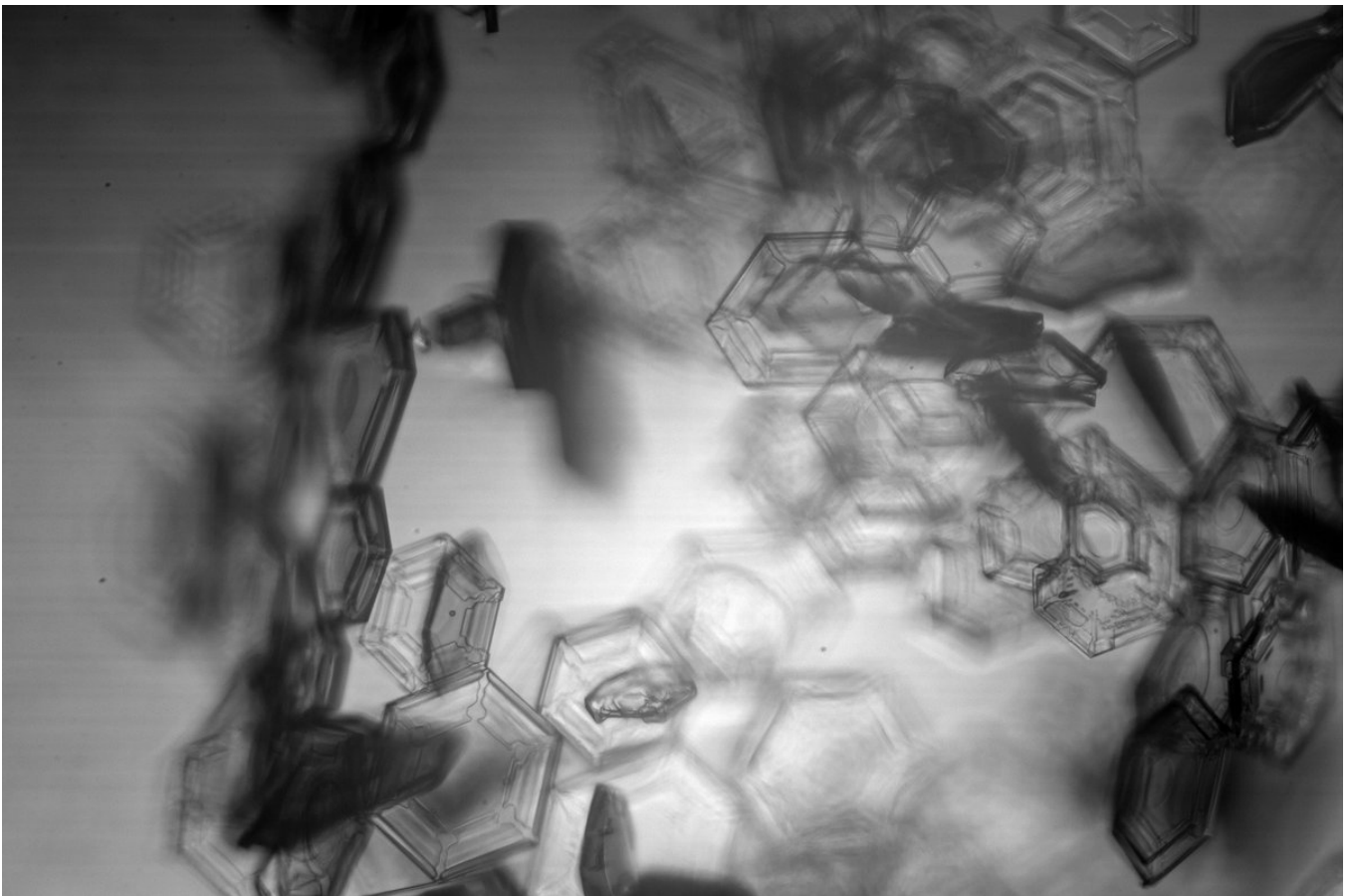
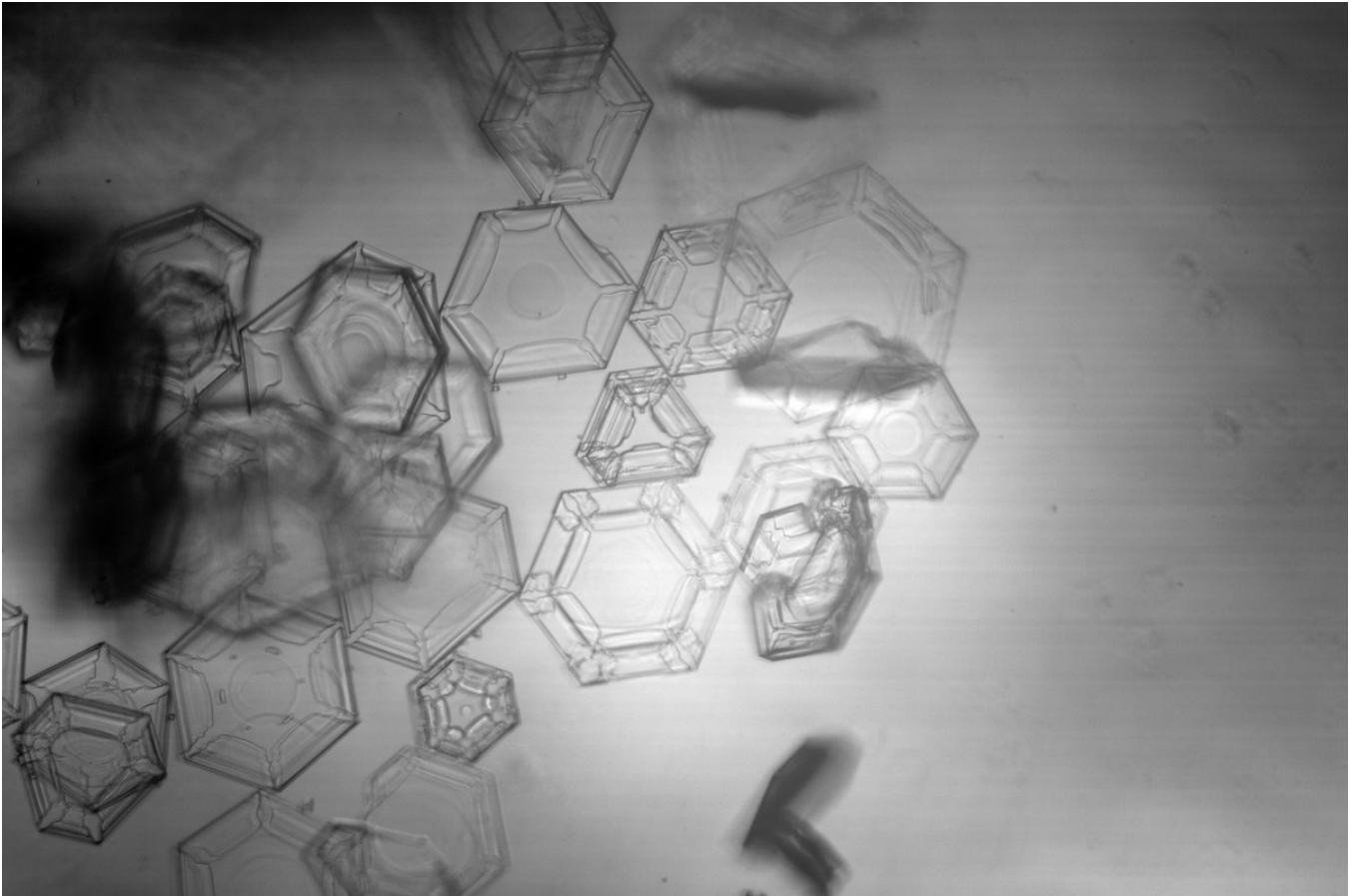


5/6 - 2023-12-03 19:58 UTC

2023 Nov 20/21 - stage 3 - 5/6. More crystals. But are we now any the wiser on how these subparhelic arcs form? And why only the outer arcs? Seeking for that answer from the crystal sample is compounded by the fact that the display has also parhelia, subparhelia and parhelic circle, meaning not all of those crystals caused the subparhelic arcs, maybe just a minor proportion. So let's wait for that display that has only the subparhelic arcs. It is not fantasy, such a display has been photographed: <https://t.co/89BDgSwQD9>

And one more thing. The presence of parhelic circle makes the absence of 46 degree halos even more strange.





6/6 - 2023-12-03 20:01 UTC

2023 Nov 20/21 - stage 3 - 6/6. More crystals. After I had finished the crystal photography ice fog had

disappeared completely and it was time to go look for it across the river at Teollisuuskylä (which translates to Industrial Village).

